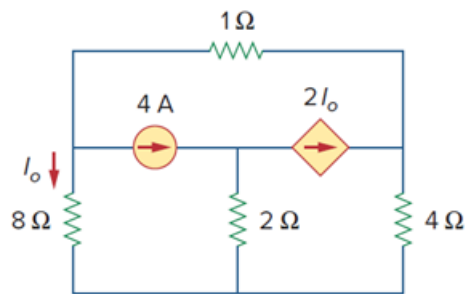


Problem

Find I_o in the circuit of Fig. 3.59.

Figure 3.59:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.59 in the textbook.

Draw the modified circuit diagram with node voltages.

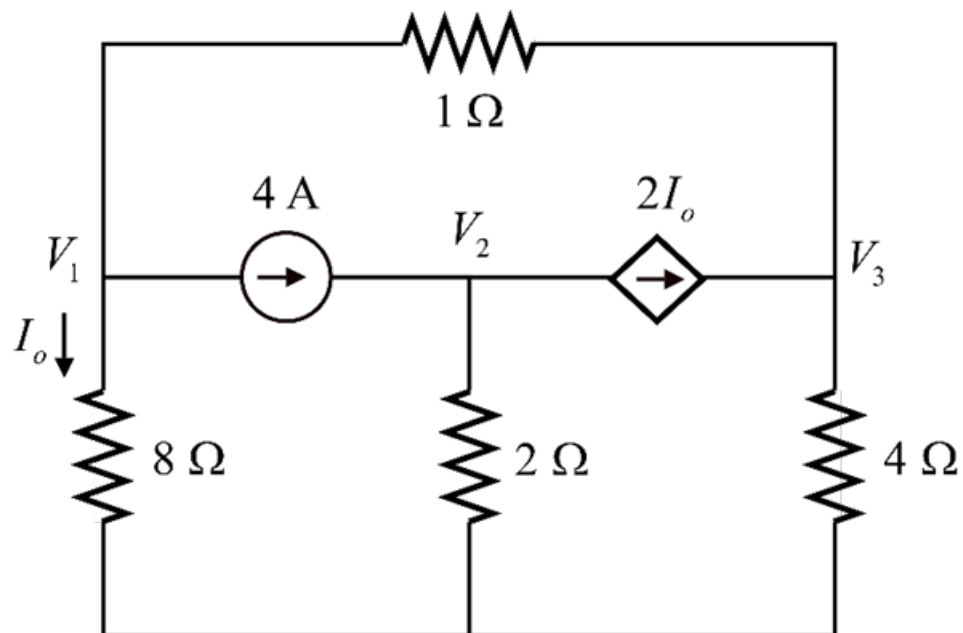


Figure 1

Step 2 of 3

From the Figure,

$$I_o = \frac{V_1}{8}$$

$$V_1 = 8I_o$$

Apply nodal analysis at the node V_1 .

$$\frac{V_1 - V_3}{1} + I_o + 4 = 0$$

$$V_1 - V_3 + I_o + 4 = 0$$

Substitute $8I_o$ for V_1 .

$$8I_o - V_3 + I_o + 4 = 0$$

$$V_3 = 9I_o + 4$$

Step 3 of 3

Apply nodal analysis at the node V_3 .

$$\frac{V_3}{4} + \frac{V_3 - V_1}{1} = 2I_o$$

$$\frac{V_3 + 4(V_3 - V_1)}{4} = 2I_o$$

$$5V_3 - 4V_1 = 8I_o$$

Substitute $8I_o$ for V_1 and $9I_o + 4$ for V_3 .

$$5(9I_o + 4) - 4(8I_o) = 8I_o$$

$$45I_o + 20 - 32I_o = 8I_o$$

$$(8 - 45 + 32)I_o = 20$$

$$-5I_o = 20$$

$$I_o = \frac{20}{-5}$$

$$= -4 \text{ A}$$

Thus, the value of output current, I_o is -4 A.